

Exercices on session *Working with data*

Working with data from the *Inventário Florestal Nacional*

- Go to the website of the IFN ([here](#)) and download the data of the Rondônia for trees with a DBH ≥ 10 cm (don't forget to look at the metadata!)
- Import these data into R (make sure you look at the data structure before)

```
library(readxl)
data_Ron <- read_excel(path = "data/ifn_dap10_rondonia_disp-set2025.xlsx")
```

- How many variables are there? how many observations?

```
ncol(data_Ron)
```

```
[1] 44
```

```
nrow(data_Ron)
```

```
[1] 31133
```

```
dim(data_Ron)
```

```
[1] 31133    44
```

- Look at the structure, everything is fine?

```
str(data_Ron)
```

```

tibble [31,133 x 44] (S3: tbl_df/tbl/data.frame)
 $ Lotes           : chr [1:31133] "RO-01" "RO-01" "RO-01" "RO-01" ...
 $ UA              : chr [1:31133] "RO_10" "RO_480" "RO_603" "RO_603" ...
 $ bioma           : chr [1:31133] "Amazônia" "Amazônia" "Amazônia" "Amazônia" ...
 $ uf              : chr [1:31133] "Rondônia" "Rondônia" "Rondônia" "Rondônia" ...
 $ mun             : chr [1:31133] "Porto Velho" "Costa Marques" "Pimenteiras do Oeste" ...
 $ lon_pc          : num [1:31133] -63 -64.4 -61.6 -61.6 -62.8 ...
 $ lat_pc          : num [1:31133] -8.28 -12.24 -13.5 -13.5 -10.8 ...
 $ Subunidade      : num [1:31133] 2 3 1 1 3 3 3 3 3 3 ...
 $ Subparcela      : num [1:31133] 5 1 1 5 2 2 2 2 2 2 ...
 $ Data            : chr [1:31133] "01/10/2015" "05/11/2015" "22/11/2015" "22/11/2015" ...
 $ Sem_arvores_na_UA : chr [1:31133] "N" "N" "N" "N" ...
 $ Narv            : num [1:31133] 46 9 1 16 3 3 3 3 3 3 ...
 $ Nfuste          : num [1:31133] 1 1 1 1 4 8 5 6 7 1 ...
 $ Codigo_medicao   : chr [1:31133] "RO_10_2_5_46" "RO_480_3_1_9" "RO_603_1_1_1" "RO_603_1_1_1" ...
 $ Especie_campo   : chr [1:31133] "buiçu" "buiçu" "buiçu" "buiçu" ...
 $ Codigo_Coleta   : chr [1:31133] "RO_10_2_5_46" "RO_10_2_5_46" "RO_10_2_5_46" "RO_10_2_5_46" ...
 $ Nivel           : logi [1:31133] NA NA NA NA NA NA ...
 $ DAP             : num [1:31133] 24 51.5 34.1 18.8 11.5 10.3 10.3 14 27.2 20.5 ...
 $ DB1             : logi [1:31133] NA NA NA NA NA NA ...
 $ DB2            : logi [1:31133] NA NA NA NA NA NA ...
 $ SA              : num [1:31133] 1 1 1 1 1 1 1 1 1 1 ...
 $ QF              : num [1:31133] 2 1 2 2 3 2 2 2 1 2 ...
 $ PS              : num [1:31133] 2 2 3 3 3 3 3 3 3 3 ...
 $ AFF            : chr [1:31133] "N" "N" "N" "N" ...
 $ HT              : num [1:31133] 18 20 10.7 12 8.3 7.9 7.5 7.8 9.6 8.1 ...
 $ HF              : num [1:31133] 10 12 4.3 6 5.6 4.2 2.2 2.3 2.2 5.2 ...
 $ PL              : chr [1:31133] "S" "N" "N" "N" ...
 $ HM              : chr [1:31133] "N" "N" "S" "N" ...
 $ HAB            : logi [1:31133] NA NA NA NA NA NA ...
 $ Obs            : chr [1:31133] NA NA NA NA ...
 $ Tocos          : num [1:31133] 0 0 0 0 0 0 0 0 0 0 ...
 $ MB              : chr [1:31133] "S" "N" "N" "N" ...
 $ DAP_GRUPO       : chr [1:31133] "DAP10" "DAP10" "DAP10" "DAP10" ...
 $ verbatimIdentification: chr [1:31133] "Abarema sp. 3" "Abarema sp. 3" "Abarema sp. 3" "Abarema sp. 3" ...
 $ id_FFB          : num [1:31133] 22756 22756 22756 22756 22756 ...
 $ id_POWO         : logi [1:31133] NA NA NA NA NA NA ...
 $ family          : chr [1:31133] "Fabaceae" "Fabaceae" "Fabaceae" "Fabaceae" ...
 $ genus           : chr [1:31133] "Abarema" "Abarema" "Abarema" "Abarema" ...
 $ specificEpithet : chr [1:31133] NA NA NA NA ...
 $ infraspecificEpithet : logi [1:31133] NA NA NA NA NA NA ...
 $ scientificName   : chr [1:31133] "Abarema Pittier" "Abarema Pittier" "Abarema Pittier" "Abarema Pittier" ...
 $ taxonRank        : chr [1:31133] "GENERO" "GENERO" "GENERO" "GENERO" ...

```

```
$ venacularName      : logi [1:31133] NA NA NA NA NA NA ...
$ Codigo_coleta      : logi [1:31133] NA NA NA NA NA NA ...
```

- Transform the following variable into factors: *Unidade de amostragem*, *Sanidade do indivíduo*, *Qualidade do fuste do indivíduo*, *Posição sociológica*, *Hábito dos indivíduos* and check the structure again

```
data_Ron$UA <- as.factor(data_Ron$UA) # Unidade de amostragem
data_Ron$SA <- as.factor(data_Ron$SA) # Sanidade do indivíduo
data_Ron$QF <- as.factor(data_Ron$QF) # Qualidade do fuste do indivíduo
data_Ron$PS <- as.factor(data_Ron$PS) # Posição sociológica
data_Ron$HAB <- as.factor(data_Ron$HAB) # Hábito dos indivíduos
```

- Transform the following variable into booleans: *Arvore fora da floresta*, *Presença de Lianas*, *Coleta de material botânico* and check the structure again

```
data_Ron$AFF <- as.logical(data_Ron$AFF) # Arvore fora da floresta
data_Ron$PL <- as.logical(data_Ron$PL) # Presença de Lianas
data_Ron$MB <- as.logical(data_Ron$MB) # Coleta de material botânico
str(data_Ron)
```

```
tibble [31,133 x 44] (S3: tbl_df/tbl/data.frame)
 $ Lotes      : chr [1:31133] "RO-01" "RO-01" "RO-01" "RO-01" ...
 $ UA         : Factor w/ 286 levels "RO_1","RO_10",...: 2 198 255 255 110 110 110 ...
 $ bioma      : chr [1:31133] "Amazônia" "Amazônia" "Amazônia" "Amazônia" ...
 $ uf         : chr [1:31133] "Rondônia" "Rondônia" "Rondônia" "Rondônia" ...
 $ mun        : chr [1:31133] "Porto Velho" "Costa Marques" "Pimenteiras do Oeste" ...
 $ lon_pc     : num [1:31133] -63 -64.4 -61.6 -61.6 -62.8 ...
 $ lat_pc     : num [1:31133] -8.28 -12.24 -13.5 -13.5 -10.8 ...
 $ Subunidade : num [1:31133] 2 3 1 1 3 3 3 3 3 3 ...
 $ Subparcela : num [1:31133] 5 1 1 5 2 2 2 2 2 2 ...
 $ Data       : chr [1:31133] "01/10/2015" "05/11/2015" "22/11/2015" "22/11/2015" ...
 $ Sem_arvores_na_UA : chr [1:31133] "N" "N" "N" "N" ...
 $ Narv       : num [1:31133] 46 9 1 16 3 3 3 3 3 3 ...
 $ Nfuste     : num [1:31133] 1 1 1 1 4 8 5 6 7 1 ...
 $ Codigo_medicao : chr [1:31133] "RO_10_2_5_46" "RO_480_3_1_9" "RO_603_1_1_1" "RO_603_1_1_1" ...
 $ Especie_campo : chr [1:31133] "buiçu" "buiçu" "buiçu" "buiçu" ...
 $ Codigo_Coleta : chr [1:31133] "RO_10_2_5_46" "RO_10_2_5_46" "RO_10_2_5_46" "RO_10_2_5_46" ...
 $ Nivel      : logi [1:31133] NA NA NA NA NA NA ...
 $ DAP        : num [1:31133] 24 51.5 34.1 18.8 11.5 10.3 10.3 14 27.2 20.5 ...
 $ DB1        : logi [1:31133] NA NA NA NA NA NA ...
 $ DB2        : logi [1:31133] NA NA NA NA NA NA ...
```

```

$ SA : Factor w/ 4 levels "1","2","3","4": 1 1 1 1 1 1 1 1 1 1 ...
$ QF : Factor w/ 4 levels "1","2","3","4": 2 1 2 2 3 2 2 2 1 2 ...
$ PS : Factor w/ 4 levels "1","2","3","4": 2 2 3 3 3 3 3 3 3 3 ...
$ AFF : logi [1:31133] NA NA NA NA NA NA NA ...
$ HT : num [1:31133] 18 20 10.7 12 8.3 7.9 7.5 7.8 9.6 8.1 ...
$ HF : num [1:31133] 10 12 4.3 6 5.6 4.2 2.2 2.3 2.2 5.2 ...
$ PL : logi [1:31133] NA NA NA NA NA NA NA ...
$ HM : chr [1:31133] "N" "N" "S" "N" ...
$ HAB : Factor w/ 0 levels: NA NA NA NA NA NA NA NA NA NA ...
$ Obs : chr [1:31133] NA NA NA NA ...
$ Tocos : num [1:31133] 0 0 0 0 0 0 0 0 0 0 ...
$ MB : logi [1:31133] NA NA NA NA NA NA NA ...
$ DAP_GRUPO : chr [1:31133] "DAP10" "DAP10" "DAP10" "DAP10" ...
$ verbatimIdentification: chr [1:31133] "Abarema sp. 3" "Abarema sp. 3" "Abarema sp. 3" "Abarema sp. 3" ...
$ id_FFB : num [1:31133] 22756 22756 22756 22756 22756 ...
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$ family : chr [1:31133] "Fabaceae" "Fabaceae" "Fabaceae" "Fabaceae" ...
$ genus : chr [1:31133] "Abarema" "Abarema" "Abarema" "Abarema" ...
$ specificEpithet : chr [1:31133] NA NA NA NA ...
$ infraspecificEpithet : logi [1:31133] NA NA NA NA NA NA NA ...
$ scientificName : chr [1:31133] "Abarema Pittier" "Abarema Pittier" "Abarema Pittier" "Abarema Pittier" ...
$ taxonRank : chr [1:31133] "GENERO" "GENERO" "GENERO" "GENERO" ...
$ venacularName : logi [1:31133] NA NA NA NA NA NA NA ...
$ Codigo_coleta : logi [1:31133] NA NA NA NA NA NA NA ...

```

- How many *Unidade amostral* is there?

```
length(levels(data_Ron$UA))
```

```
[1] 286
```

```
# alternative (but doesn't allow storing the result)
str(data_Ron$UA)
```

```
Factor w/ 286 levels "RO_1","RO_10",...: 2 198 255 255 110 110 110 110 110 110 ...
```

```
# alternative not seen in class:
length(unique(data_Ron$UA))
```

```
[1] 286
```

- Add a column with the DBH in meter

```
data_Ron$DAP_m <- data_Ron$DAP / 100
```

- What are the minimum, maximum and median height (*Altura total*)

```
min(data_Ron$HT)
```

```
[1] 1
```

```
max(data_Ron$HT)
```

```
[1] 53.5
```

```
# or
range(data_Ron$HT)
```

```
[1] 1.0 53.5
```

```
median(data_Ron$HT)
```

```
[1] 14
```

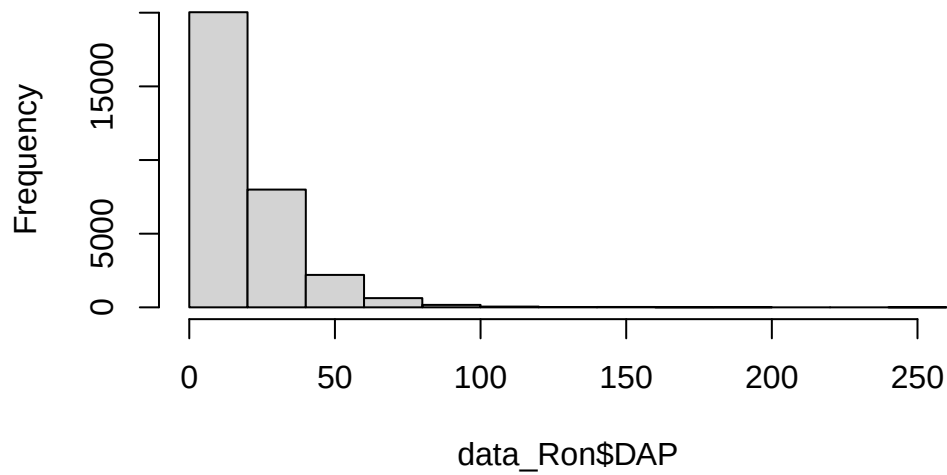
```
# or
summary(data_Ron$HT)
```

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1.00	10.50	14.00	14.63	18.00	53.50

- Represent the distribution of DBH (in cm) using an histogram (*Bonus question for the foresters and forest ecologists: does this distribution make sense to you?*)

```
hist(data_Ron$DAP)
```

Histogram of data_Ron\$DAP



- How many trees are dead standing (*Sanidade do indivíduo* indicates *morta em pé*)?

```
table(data_Ron$SA)
```

```
  1    2    3    4
25686 2452  416 2579
```

```
# alternative not seen in class
sum(data_Ron$SA == 4)
```

```
[1] 2579
```

```
# or
dim(data_Ron[data_Ron$SA == 4,])[1]
```

```
[1] 2579
```

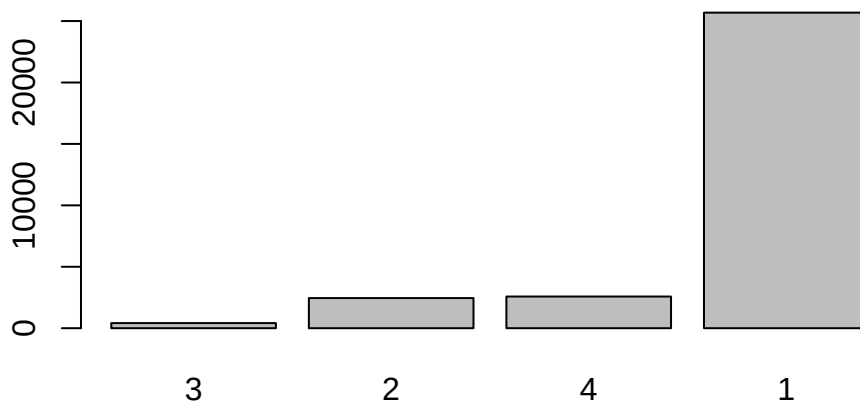
- Which percentage of trees are dead standing?

```
questionr::freq(data_Ron$SA)
```

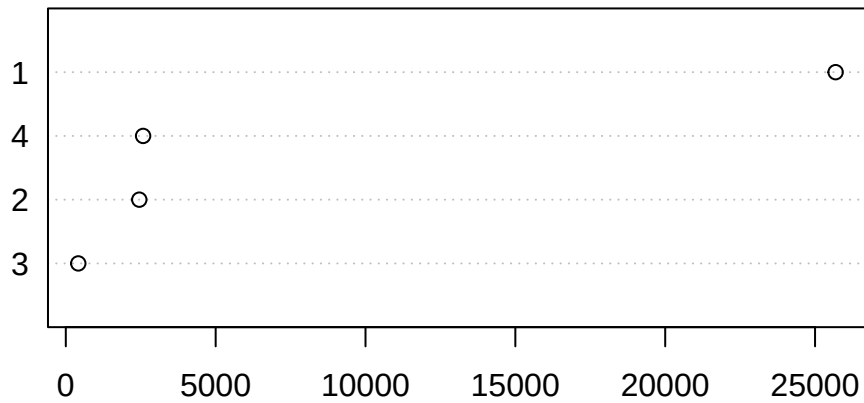
	n	%	val%
1	25686	82.5	82.5
2	2452	7.9	7.9
3	416	1.3	1.3
4	2579	8.3	8.3

- Make a graphical representation (barplot or Cleveland plot) of the occurrences of health levels (*Sanidade do indivíduo*)

```
tab <- table(data_Ron$SA)
tab <- sort(tab) # order in ascending order
barplot(tab)
```



```
tab <- table(data_Ron$SA)
tab <- sort(tab) # order in ascending order
dotchart(as.vector(tab), # convert the table to a vector
         labels = names(tab))
```



Find the problems

We are working with a fake data set of vegetation survey in the DF, that contain the following variables:

- *site*: name of the site
- *plot*: plot ID (unique identifier)
- *height_max*: height of the tallest tree on the plot (in m)
- *DBH_max*: DBH (diameter at breast height) of the largest tree on the plot (in cm)

You can find this data set [here](#).

There are 4 problems in this data frame, what are they?

```
str(dt_pb)
```

```
'data.frame':  5 obs. of  4 variables:
 $ site      : chr  "Brasilia" "Brasilia" "brasilia" "Taguatinga" ...
 $ plot      : num   1  2  3  4  4
 $ height_max: chr   "10.3" "8.5" "9.6" "4,7" ...
 $ DBH_max   : num   20 15 31 888 12
```

```
summary(dt_pb)
```

	site	plot	height_max	DBH_max
Length	: 5	Min. :1.0	Length :5	Min. : 12.0
N.unique	: 3	1st Qu.:2.0	N.unique :5	1st Qu.: 15.0
N.blank	: 0	Median :3.0	N.blank :0	Median : 20.0
Min.nchar	: 8	Mean :2.8	Min.nchar:3	Mean :193.2
Max.nchar	:10	3rd Qu.:4.0	Max.nchar:4	3rd Qu.: 31.0
		Max. :4.0		Max. :888.0

```
# There is a uncorrect decimal separator in height_max
mode(dt_pb$height_max)
```

```
[1] "character"
```

```
dt_pb$height_max
```

```
[1] "10.3" "8.5" "9.6" "4,7" "9.8"
```

```
# There is a problem of encoding in site
dt_pb$site
```

```
[1] "Brasilia" "Brasilia" "brasilia" "Taguatinga" "Taguatinga"
```

```
# or better:
dt_pb$site <- as.factor(dt_pb$site)
levels(dt_pb$site)
```

```
[1] "brasilia" "Brasilia" "Taguatinga"
```

```
# or
table(dt_pb$site)
```

brasilia	Brasilia	Taguatinga
1	2	2

```
# There is an incoherent value in DBH_max  
summary(dt_pb$DBH_max)
```

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
12.0	15.0	20.0	193.2	31.0	888.0

```
# or  
range(dt_pb$DBH_max)
```

```
[1] 12 888
```

```
# There is a duplicated plot ID  
table(dt_pb$plot)
```

```
1 2 3 4  
1 1 1 2
```

```
# alternative  
length(unique(table(dt_pb$plot))) == 1
```

```
[1] FALSE
```

```
# alternative not seen in the class  
length(unique(dt_pb$plot)) == dim(dt_pb)[1]
```

```
[1] FALSE
```